

A4. Emergence of Segregation in Evolving Social Networks: A Link-Rewiring Generalization of Schelling’s Model

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1 Introduction

In 1971, Thomas Schelling introduced a famous agent-based model to explain how racial segregation occurs in cities [2]. His main finding was striking: even if individuals are relatively tolerant and only prefer a small fraction of similar neighbors, their collective, decentralized choices inevitably lead to a completely segregated society. This model is a classic example of an emergent phenomenon, showing how micro-level individual preferences create macro-level structural patterns that nobody explicitly intended.

However, the traditional Schelling model is limited by its spatial layout, which relies on a rigid two-dimensional grid or lattice. While a grid works well for mapping physical neighborhoods, it does not capture modern social interactions. Human relationships are better represented as complex networks where connectivity is fluid, irregular, and changes over time, rather than being restricted to immediate physical neighbors.

To adapt Schelling’s ideas to network science, this report focuses on a “link-rewiring” approach based on the work of Henry, Pralat, and Zhang (2011) [1]. Instead of moving agents to different empty nodes on a fixed graph, this model allows the network’s edges to evolve. If an agent is connected to someone from a different group, there is a certain probability—the aversion bias—that they will cut that connection and rewire it to a random agent of their own group.

The objective of this assignment is to implement this rewiring model in Python and analyze how segregation develops over time. Following the task requirements, we will explore the key parameters of the system, use sequential network visualizations to show the clustering process, and perform ensemble averaging over multiple simulation runs to guarantee statistically reliable results.

2 Model Generalization and Mechanism

2.1 Limitations of the Lattice Approach

The classic Schelling model places agents on a rigid two-dimensional grid where each agent can only interact with their immediate Moore neighborhood (the 8 surrounding cells). While this spatial structure is highly intuitive for geographical neighborhoods, it has significant limitations when applied to broader social dynamics.

First, a grid assumes a perfectly uniform distribution of contacts, meaning every single individual has the exact same potential capacity for relationships. In reality, human social structures are highly irregular; some individuals are “hubs” with many connections, while others are relatively isolated. Second, spatial grids imply that physical proximity is the only driver of

relationships. In a modern context, social ties are fluid and not strictly bound by geography. Moving a physical agent to a new location change all their neighbors simultaneously, which is a very drastic structural shift.

2.2 The Link-Rewiring Framework

To address these limitations, we implement the generalization proposed by Henry, Prałat, and Zhang (2011) [1], which shifts the focus from spatial movement to network adaptation. Instead of moving nodes across a static canvas, the agents remain fixed as nodes in a graph $G = (V, E)$, and the connections (edges) between them dynamically evolve over time.

We consider a population of N nodes split into two distinct groups (Group 0 and Group 1) based on a population ratio ρ . The total number of edges $|E|$ in the network is kept strictly constant throughout the simulation to maintain a stable network density.

The discrete-time dynamics of the model follow a strict algorithmic loop at each time step:

1. **Identification:** The system identifies all active "cross-edges" in the network. A cross-edge is defined as any link $(u, v) \in E$ where the attribute of node u is different from the attribute of node v (i.e., they belong to different groups).
2. **Selection:** One cross-edge (u, v) is selected uniformly at random from the pool of active cross-edges.
3. **Pruning (Aversion Bias):** With a defined probability p (the aversion bias parameter), the selected edge is deleted from the network:

$$E \leftarrow E \setminus \{(u, v)\} \quad (1)$$

4. **Biased Rewiring:** To maintain a constant edge count, the dissatisfied node u immediately seeks a new connection. Node u randomly selects a node w from the network that satisfies two strict conditions:

- Node w belongs to the same group as node u .
- There is no existing edge between node u and node w ($(u, w) \notin E$).

Once found, the new internal edge is established: $E \leftarrow E \cup \{(u, w)\}$.

2.3 Quantifying Segregation

To track the emergence of segregation at any point during the simulation, we calculate the global Homophily Index (H). This metric tracks the total fraction of edges that connect nodes belonging to the exact same group:

$$H = \frac{E_{\text{internal}}}{E_{\text{total}}} = \frac{|\{(u, v) \in E : \text{type}(u) = \text{type}(v)\}|}{|E|} \quad (2)$$

When the network is initialized as a completely random graph, the homophily index floats around $H \approx 0.5$ (assuming a 50/50 group split). As the rewiring process drives the network toward complete segregation, the index approaches an absorbing steady state of $H = 1.0$, where cross-group ties are entirely eliminated.

3 Experimental Setup and System Parameters

To systematically evaluate the link-rewiring generalization of Schelling’s model, we developed a computational framework in Python. Because this model relies on stochastic rules—such as selecting random edges and choosing random compatible partners—the configuration of the initial parameters determines the trajectory, speed, and final structural state of the network.

Below is the explanation of each relevant parameter, its role in the system, and the specific values selected for our experiments:

- **Network Size (N):** This variable sets the total population size of agents within our closed system. For all simulations, we fix this value at $N = 200$. This size is small enough to keep computation efficient while being large enough to form distinct, statistically observable structural clusters.
- **Initial Average Degree (K):** This parameter determines the baseline connectivity density of the network. We set $K = 6$, meaning that before any rewiring takes place, each agent has an average of 6 social connections. This provides a moderately connected graph where agents have enough links to evaluate diversity without being oversaturated.
- **Aversion Bias (p):** This is the behavioral parameter representing the probability that an agent will choose to break an active tie with a neighbor from a different group. By default, we set a baseline value of $p = 0.4$. To test the network’s phase transitions, we vary this value in later experiments using $p = 0.02$ (high tolerance) and $p = 0.95$ (high intolerance).
- **Population Balance (ρ):** This controls the demographic ratio between the two groups. Our baseline model utilizes a perfectly balanced split of $\rho = 0.5$ (50% Group 0 and 50% Group 1). To observe how minority groups behave under segregation pressures, we also test an asymmetric scenario with $\rho = 0.2$ (a 20/80 minority-majority split).
- **Initial Network Topology:** This defines the starting layout of social ties. Our default topology is an Erdős-Rényi (ER) random graph where edges are placed with a uniform probability $p_{edge} = K/(N - 1)$. As a comparative experiment, we replace this with a scale-free Barabási-Albert (BA) network structure to study how highly connected “hub” nodes change the speed of segregation.
- **Simulation Length (Iterations):** This defines the maximum discrete time steps allowed for the evolution. We set a maximum ceiling of 5000 iterations, which ensures the network has plenty of time to find an absorbing steady state or plateau before the script ends.

3.1 Summary of Experimental Runs

To thoroughly analyze the system, we organized our simulation results into seven distinct setups using the specific parameter sets detailed below:

3.2 Ensemble Averaging Protocol

Because the simulation updates links by randomly sampling active cross-edges, a single simulation run is highly vulnerable to random noise or statistical outliers. A single “lucky” sequence of random selections can cause the system to converge artificially fast or get trapped in temporary local states that do not represent the true average physics of the model.

To filter out this stochastic noise and ensure scientific reliability, we implemented an ensemble averaging protocol. For every parameter set evaluated, the simulation loop is repeated

Table 1: Parameter Matrix for Baseline and Exploratory Simulations

Simulation Type	Nodes (N)	Avg. Degree (K)	Aversion (p)	Balance (ρ)	Graph Topology
1. Baseline Control	200	6	0.40	0.5	Erdős-Rényi (ER)
2. Highly Tolerant	200	6	0.02	0.5	Erdős-Rényi (ER)
3. Highly Intolerant	200	6	0.95	0.5	Erdős-Rényi (ER)
4. Asymmetric Minority	200	6	0.40	0.2	Erdős-Rényi (ER)
5. Scale-Free Structure	200	6	0.40	0.5	Barabási-Albert (BA)
6. Low Average Degree	200	2	0.40	0.5	Erdős-Rényi (ER)
7. High Average Degree	200	20	0.40	0.5	Erdős-Rényi (ER)

across 20 independent runs utilizing identical variables but unique random number seeds. During each run, the global Homophily Index is recorded at regular intervals. We then aggregate these arrays to compute the mathematical mean (μ) for the time-series curves and calculate the standard deviation (σ). In our figures, this variance is represented as a shaded error envelope surrounding the mean path, ensuring that our final interpretations reflect robust, repeatable behaviors.

4 Results and Simulation Metrics

This section presents the empirical findings obtained from the seven distinct simulation configurations outlined in our experimental design. To track the microscopic evolution of our network into macroscopic segregated structures, we analyze two primary data formats generated by our pipeline: the time-series history of the global Homophily Index (H) evaluated over an ensemble of 20 independent repetitions, and standalone network visualizations capturing the structural layouts at convergence.

To provide a comprehensive overview of the system’s kinetic behaviors, the absolute convergence times (the exact step where the system eliminates its final cross-group edge and reaches the absorbing state of $H = 1.0000$) across all configurations are compiled in Table 2.

Table 2: Empirical Convergence Times to Full Segregation ($H = 1.0$)

Case Number	Simulation Configuration Details	Convergence Step
Case 1	Baseline Control	737
Case 2 (Standard)	Highly Tolerant (5k Step Limit)	Did not converge ($H = 0.6667$)
Case 2 (Extended)	Highly Tolerant (50k Step Limit)	
Case 3	Highly Intolerant	326
Case 4	Asymmetric Minority	535
Case 5	Scale-Free Structure ($m = 3$, BA)	724
Case 6	Low Average Degree	323
Case 7	High Average Degree	2,370

4.1 The Baseline Control Profile (Case 1)

The baseline control simulation provides a point of comparison for the system’s core dynamics. Initialized as a uniformly mixed Erdős-Rényi graph, the system begins near a perfectly integrated state of $H \approx 0.50$. Under an aversion bias of $p = 0.40$, cross-group links are steadily dismantled and rewired.

As shown in Figure 1, the Homophily Index follows a linear, ascendant trajectory during the early phase before smoothing into a plateau as it hits $H = 1.0000$ at step 737. The final network configuration reveals a complete topological phase transition. Because links crossing

the two demographics are irreversibly broken, the single connected giant component splits apart into two completely independent, monochromatic topological islands.

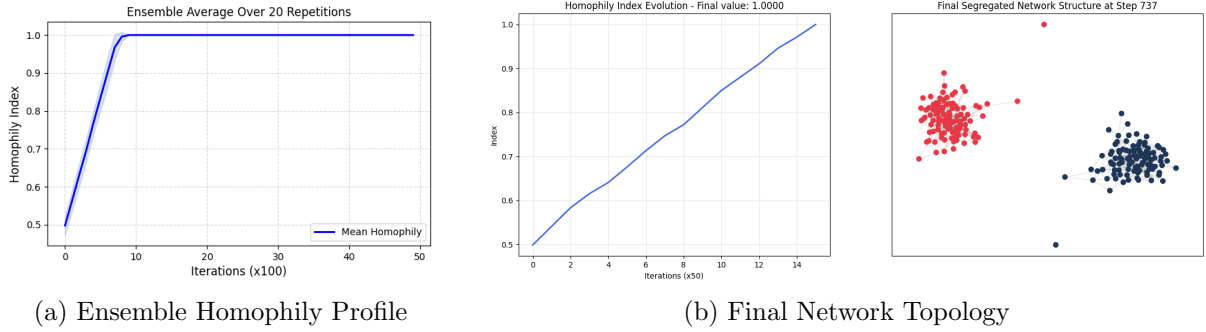


Figure 1: Case 1 - Baseline Control

4.2 The Kinetic Role of Aversion Bias (Cases 2 and 3)

Adjusting the behavioral parameter p uncovers a powerful relationship between individual behavioral tolerance and global convergence velocity.

In the highly tolerant scenario (Case 2), set at $p = 0.02$, individual nodes are highly willing to maintain diverse social links. When restricted to a standard window of 5,000 iterations, the system fails to achieve total segregation, ending at a partial homophily value of $H = 0.6667$ (Figure 2). Visually, the network layout preserves an integrated structure, functioning as a single connected component with overlapping group boundaries.

However, extending the simulation ceiling to 50,000 iterations reveals an essential property of this model: the system eventually achieves complete segregation ($H = 1.0000$) at step 16,874, fracturing into two distinct topological components as shown in Figure 3. This experiment provides clear proof of the "absorbing state" theory outlined by Henry et al. (2011). Because the rewiring rule works like a one-way valve—destroying cross-edges with a low probability but creating internal edges with absolute certainty—any bias greater than zero ($p > 0$) acts as a relentless structural pressure that forces the system into complete segregation given enough time.

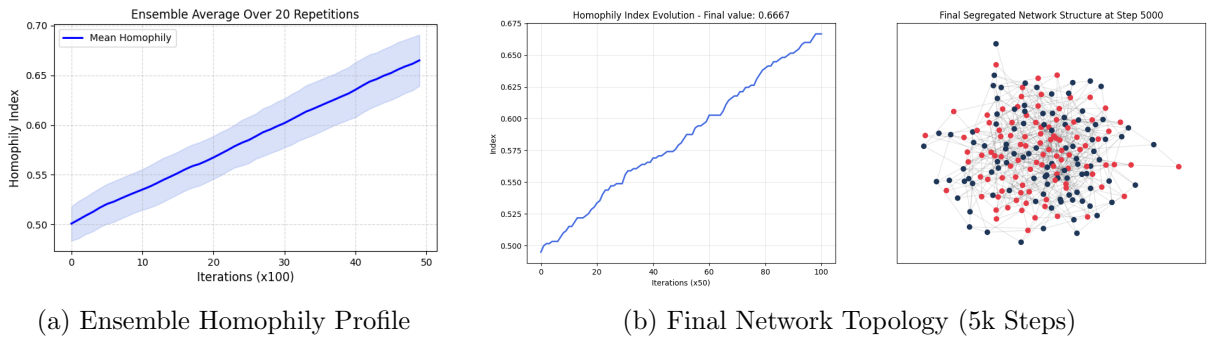


Figure 2: Case 2 (Standard) - Highly Tolerant

Inversely, Case 3 ($p = 0.95$) represents a highly intolerant population. Here, the cross-group link elimination mechanism acts rapidly. As seen in Figure 4, the network experiences rapid sorting, driving the homophily index to its maximum threshold almost immediately and achieving absolute segregation at step 326.

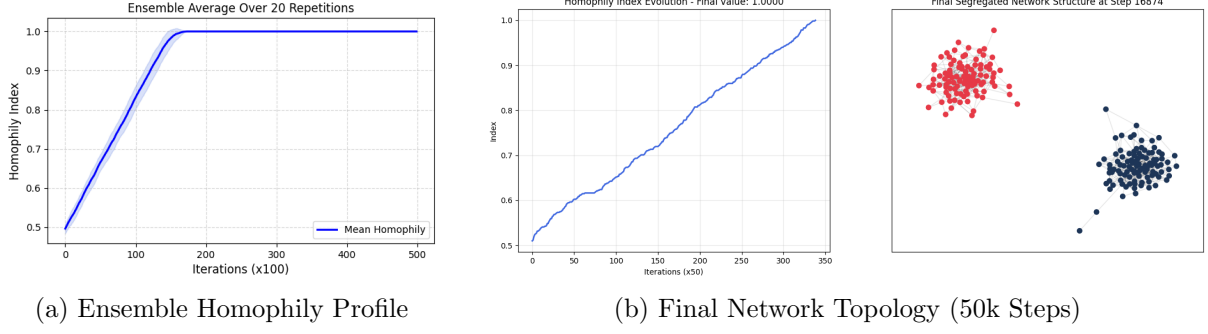


Figure 3: Case 2 (Extended) - Highly Tolerant

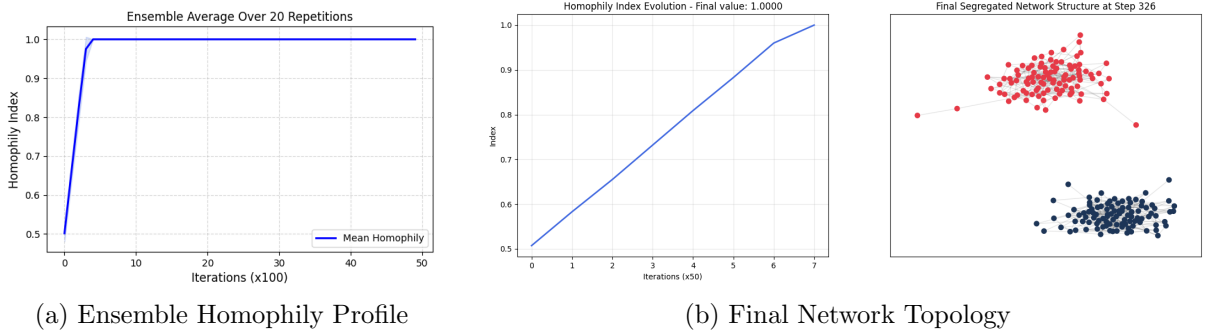


Figure 4: Case 3 - Highly Intolerant

4.3 Demographic Asymmetry and the Structural Head-Start (Case 4)

Case 4 explores a minority-majority social structure by imposing an uneven population distribution of $\rho = 0.20$ (a 20/80 split). This configuration alters the system's initial conditions. In a perfectly mixed random graph with an uneven population balance, the probability that a randomly placed edge connects two nodes of the same type can be calculated mathematically as:

$$H_{\text{initial}} = \rho^2 + (1 - \rho)^2 = (0.20)^2 + (0.80)^2 = 0.04 + 0.64 = 0.68 \quad (3)$$

Empirically, our simulation confirms this behavior: the ensemble time series (Figure 5) begins with an inherent structural "head-start" at $H \approx 0.67$. Because the system starts with a significantly smaller pool of active cross-group links to clear out, it converges much faster than the 50/50 baseline, reaching full segregation at step 535. The resulting layout shows two fully separated components where the minority group forms a smaller, distinct community next to a much larger majority network.

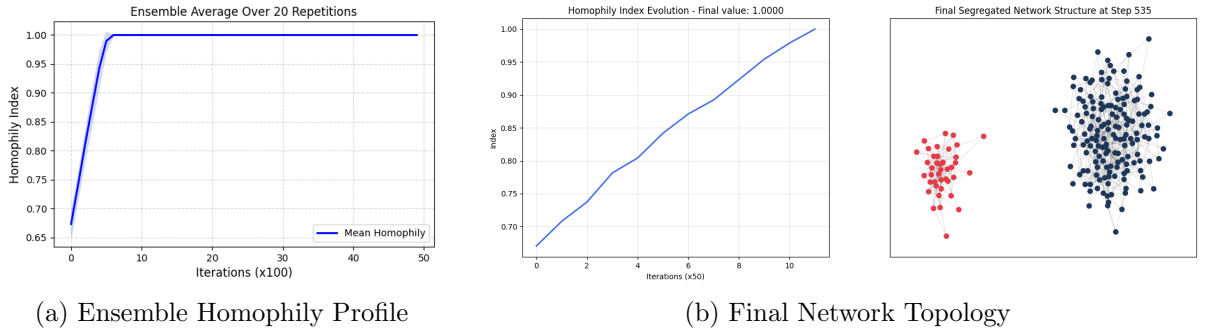


Figure 5: Case 4 - Asymmetric Minority

4.4 Topological Effects: Scale-Free vs. Random Connectivity (Case 5)

In Case 5, we replace the Erdős-Rényi initialization with a scale-free Barabási-Albert graph while maintaining a comparable average connectivity ($\langle k \rangle \approx 6$). The empirical results reveal that the scale-free topology converges slightly faster than its random graph counterpart, reaching a stable segregated state at step 724 compared to 737 steps for Case 1 (Figure 6).

Notably, this numerical difference is remarkably small (less than 15 steps), indicating that while the hub structure alters the local paths of link sorting, it does not fundamentally accelerate the global evolutionary timeline under a uniform random edge-selection protocol. The hubs belonging to Group 0 rapidly shed their connections to Group 1 and fill those vacancies with same-type neighbors. This creates localized star-like clusters that pull the rest of the network into uniform communities, but the global rate remains tightly bounded by the overarching aversion value.

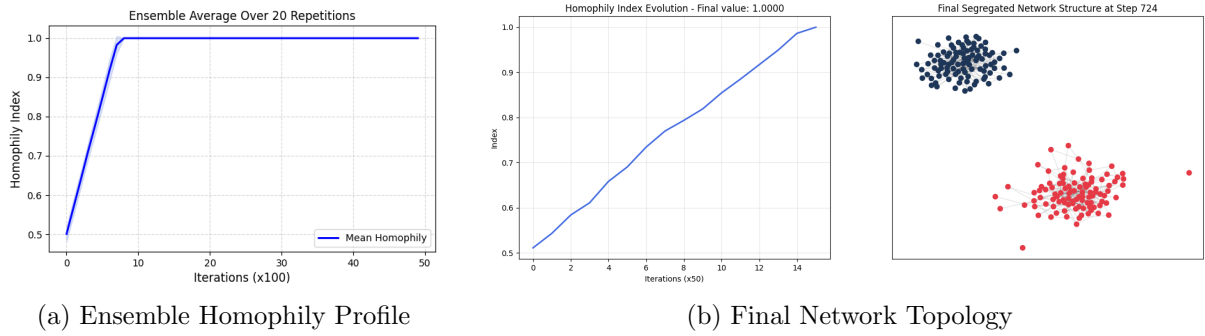


Figure 6: Case 5 - Scale-free structure

4.5 Effect of Network Density (Cases 6 and 7)

Altering the network density via the average degree parameter (K) leads to clear differences in how fast the system segregates.

Case 6 evaluates a highly sparse graph ($K = 2$). With very few active connections distributed across the system, the total volume of cross-group links is exceptionally low. Consequently, the edge removal process rapidly isolates the few conflicting ties, causing the network to hit complete segregation at step 323, making it the fastest convergence observed among all seven configurations. As observed in Figure 7, the graph does not display two clean consolidated islands, but rather fractures into a collection of scattered isolated sub-graphs due to its original structural sparsity.

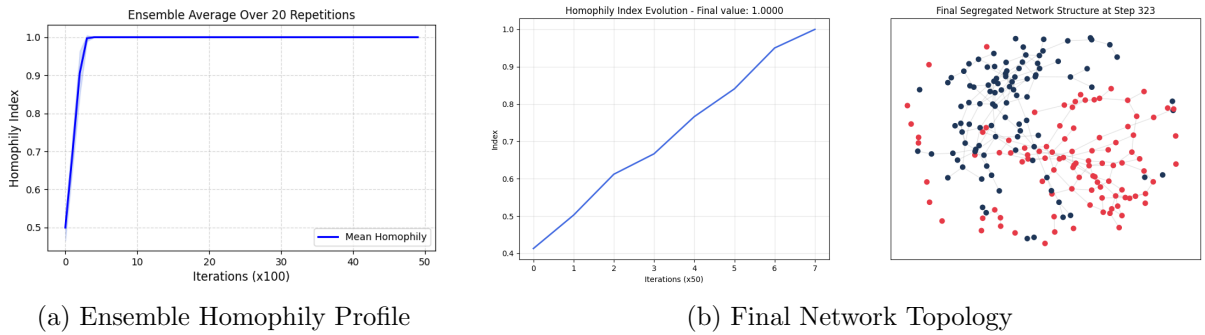


Figure 7: Case 6 - Low Average Degree

Case 7 increases connectivity to a dense configuration of $K = 20$. This structure introduces a major bottleneck, delaying full convergence until step 2,370 (Figure 8). This delay results from

a structural "crowding effect." Because every node is heavily linked, a dissatisfied agent trying to rewire an edge faces a highly saturated local environment. It becomes statistically difficult for a node to randomly locate a same-type partner that it is not already directly connected to. This lack of available partner slots creates an evolutionary bottleneck, forcing the system to run through thousands of redundant iterations before clearing out the final cross-group links.

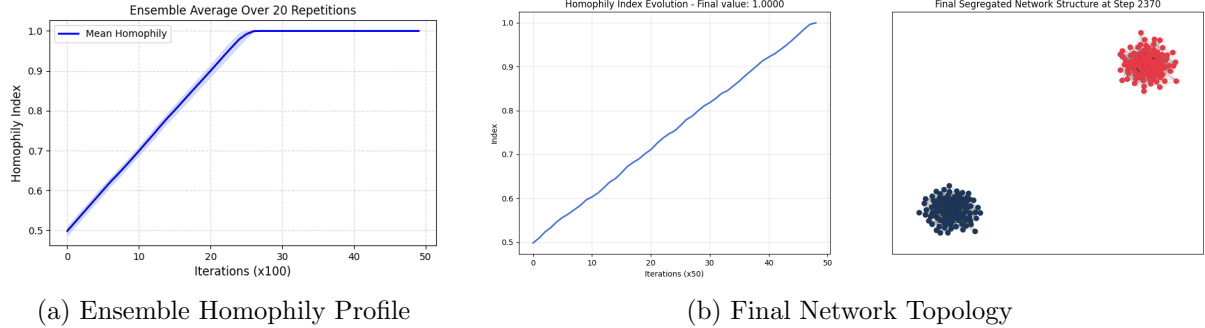


Figure 8: Case 7 - High Average Degree

5 Discussion: Emergent Phenomena and Analytical Insights

The results of our simulations show how simple, local rewiring rules create clear macroscopic changes in network structure. By evaluating seven different cases, we can analyze the specific mechanisms that drive network segregation.

5.1 Absorbing States and the Inevitability of Segregation

One of the most important findings comes from comparing Case 1 ($p = 0.40$), Case 2 ($p = 0.02$), and Case 3 ($p = 0.95$). In traditional spatial models, if individuals have high tolerance, segregation can often be avoided entirely. However, in this network rewiring model, changing the aversion bias p only changes the speed of the process, not the final result.

This happens because the rewiring rule acts like a one-way valve. Cross-group edges are deleted with a probability p , but whenever a new edge is created, it is always an internal edge between nodes of the same group. Because edges can leave the cross-group category but can never return, the system has a permanent structural bias. Total segregation ($H = 1.0$) is a mathematical absorbing state. Once the network reaches a state with zero cross-group edges, the rewiring process stops completely. This explains why Case 2, despite its high tolerance ($p = 0.02$), still achieved full segregation at step 16,874 once the simulation limit was extended.

5.2 Component Fragmentation vs. Uniform Splitting

The final layout plots show that reaching $H = 1.0$ affects the network's global connectivity in different ways depending on the parameters. In the baseline control (Case 1), the network splits cleanly into two large, independent, monochromatic components (islands).

However, Case 6 ($K = 2$) shows a completely different structural phenomenon. Because the initial graph is highly sparse, it does not have enough edges to maintain a giant component for each group. Instead of splitting into two clean islands, the edge removal process causes the network to shatter into a large collection of small, isolated sub-graphs and tree-like structures. This demonstrates that network density dictates whether segregation leads to two organized communities or total social fragmentation.

5.3 Comparison: Lattice-Based Movement vs. Network Rewiring

Shifting from Schelling’s original grid to a network framework fundamentally changes the meaning of segregation:

- **In the Grid Model:** Space is fixed, and agents move physically to find a comfortable neighborhood. The density of the space remains uniform, and agents cannot choose who moves into the vacant slots near them.
- **In the Network Model:** Agents do not move; instead, they change their relationships. This effectively models how people choose to interact only with others who are similar to them, forming isolated social bubbles where individuals adjust their friend circles rather than their physical location.

Additionally, the network model introduces structural bottlenecks that do not exist on a grid. For example, in Case 7 ($K = 20$), the dense connectivity creates a crowding effect. Because nodes already hold connections to a large portion of their own group, finding a new, valid partner becomes a statistical challenge, slowing down convergence even though individuals have a standard aversion bias ($p = 0.40$).

5.4 Comparing the Initial State with Theory

We can verify our simulation methodology by looking at the initial state of Case 4 ($\rho = 0.20$). Our random graph initialization resulted in a starting homophily index of $H \approx 0.67$. This matches the theoretical expectation for a random graph with an 80/20 population split:

$$H_{\text{theoretical}} = \rho^2 + (1 - \rho)^2 = 0.2^2 + 0.8^2 = 0.04 + 0.64 = 0.68$$

Because the system naturally starts with a higher homophily index, it requires fewer rewiring steps (535 steps) to clear the remaining cross-group ties compared to the balanced 50/50 baseline (737 steps).

6 Conclusion

In this assignment, we implemented and evaluated a network-based generalization of Schelling’s segregation model, using the link-rewiring framework proposed by Henry, Prałat, and Zhang (2011). By replacing physical movement on a grid with dynamic edge adaptation on a graph, we explored how localized structural adjustments create global social divisions.

Our experimental results from the seven simulation cases highlight three main takeaways:

1. **Inevitability of Segregation:** In a dynamic rewiring network, any aversion bias greater than zero ($p > 0$) eventually forces the system into a completely segregated absorbing state. Tolerance slows down the timeline but does not change the final structural outcome.
2. **The Bottleneck of Density:** Network density heavily influences convergence speed. Sparse networks ($K = 2$) segregate rapidly because they have fewer cross-group ties to clear, while dense networks ($K = 20$) encounter a partner-seeking bottleneck that slows down the process.
3. **Structural Constraints:** While behavioral and demographic changes drastically shift convergence timelines, underlying topological shifts (such as transitioning from an Erdős-Rényi to a Barabási-Albert scale-free graph) show minimal variation in overall convergence time when edge selection is uniform.

A key limitation of this model is that agents select new partners completely at random from the entire network pool. Real-world social networks usually evolve through local mechanisms like triadic closure (connecting to a friend of a friend).

References

- [1] Henry, A. D., Pralat, P., & Zhang, C.-Q. (2011). Emergence of segregation in evolving social networks. *Proceedings of the National Academy of Sciences*, 108(21), 8605-8610.
- [2] Schelling, T. C. (1971). Dynamic models of segregation. *Journal of Mathematical Sociology*, 1(2), 143-186.